

Using encoder-only transformers for sentiment analysis

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Model

I used DistilBERT for the first dataset and BERT-Tiny for the other two (Sanh et al. 2020; Turc et al. 2019). They both come with word embeddings included, but they still need to be trained further.

DistilBERT is a smaller version of BERT, a bidirectional encoder-only transformer model released by Google. A key conceptual leap enabled by BERT and other Transformer-based models can be understood in the context of their evolution from traditional word embeddings such as Word2Vec. Word2Vec, an earlier breakthrough in distributed representations of words, learns static embeddings that map each token to a fixed vector, capturing semantic similarity and relatedness based on co-occurrence patterns. While Word2Vec advanced the field by providing low-dimensional, dense vector representations that could be leveraged for numerous NLP tasks, these embeddings are fundamentally context-independent. That is, a given word has a single, unchanging vector irrespective of its syntactic or semantic role in a sentence.

Transformers such as BERT represent a next-generation approach, moving beyond static embeddings to produce context-dependent word vectors that reflect how a word’s meaning changes with its surrounding words. Instead of relying on a fixed embedding table, BERT and similar models dynamically compute an embedding for each token on-the-fly during the forward pass. The self-attention mechanism within the Transformer’s encoder layers allows each token representation to incorporate information from every other token in the input sequence. Thus, words that have multiple meanings depending on their context—such as “bank” referring to a financial institution or a river’s edge—receive distinct and appropriately nuanced embeddings in different sentences. This context sensitivity sets Transformer-based models apart from Word2Vec-style static embeddings, enabling more accurate and flexible semantic analysis.

IMDB

The first dataset I used was the IMDB dataset from Stanford (Maas et al. 2011). It contains movie reviews along with their associated binary sentiment polarity labels. It consists of labelled 50,000 reviews. In the entire collection, no more than 30 reviews are allowed for any given movie because reviews for the same movie tend to have correlated ratings. Further, the train and test sets contain a disjoint set of movies, so no significant performance is obtained by memorizing movie-unique terms and their associated with observed labels.

Here is a sampling from that dataset:

```
pprint(pl.read_parquet("../imdb.parquet").sample(1)[0].to_dicts())
```

```
[{'label': 1,
  'text': 'I saw this important intense film tonight. Its Richard Gere and '
          'Claire's Dane's most important and best work. Gere deserves an "
          'Oscar for his fine portrayal of a man being forced into early '
          'retirement as a sex offender registrar administrator. Claire Danes '
          'is riveting as the woman Gere handpicks to replace him - a woman he '
          'tries to teach all he can while investigating one final case in '
          '"which Gere's character is convinced one of those he is charged to "
          'monitor may be holding a young girl hostage. The subject matter is '
          'shocking, sex offenders and those who monitor them, but this film '
          '"will not soon be forgotten. I know I will be haunted by Gere's "
          'portrayal for a very long time. Not since Anthony Hopkins portrayal '
          'of a serial killer has a screen portrayal terrified and so engaged '
          "me. The film opens with a shocking statistic so don't miss the "
          "opening credits. Intense and memorable. Richard Gere's finest role "
          'proving the man can act. Danes can as well and both do '
          'extraordinarily well in this often challenging film.\n'
          '\n'
          'This is a gutsy film and Gere gives a multi-layered deeply felt '
          'performance. Just give him the Oscar now...he deserves it!'}]
```

Training and testing the SimpliBERT model was done in `traintest_imdb.py`. Some of the hyperparameters held constant were:

- AdamW as the optimizer.
- 3 training epochs.
- A weight decay of 0.
- An initial learning rate of $5 * 10^{-5}$.

The hyperparameter I varied was the batch size, the number of training examples used in one step. I tried 2, 4, and 8. Training time was roughly proportional to the batch size. Performance increased significantly as the batch size increased:

Batch size	Accuracy	Training time (s)
2	0.63	890
4	0.76	1804
8	0.90	2993

Accuracy on the training set was similar, suggesting that we avoided overfitting.

Enron Spam

The second dataset I used was the Enron Spam email dataset (Metsis, Androutsopoulos, and Paliouras 2006). It contains a total of 33,716 emails, roughly evenly divided between spam and non-spam. Most emails are on the shorter side with few extending past 512 characters. This is why I chose 512 as the max length for truncating.

Here I used the BERT-Tiny model, which is quite a bit smaller and faster than DistilBERT. It has only around 4 million parameters, as compared to DistilBERT’s 67 million. Two of the hyperparameters I held steady are the batch size at 16 and number of epochs at 4. I varied the learning rate for the optimization algorithm, again AdamW. I got the best results with the greatest learning rate, $5 * 10^{-4}$. The overall results were:

Learning rate	Accuracy
5e-4	0.98
2e-5	0.96
5e-5	0.96

Accuracy on the training set was similar, suggesting that we avoided overfitting.

Emotion

The last dataset I used was one of around 400 thousand tweets classified into one of 6 labels (Saravia et al. 2018). They are sadness, joy, love, anger, fear, and surprise.

Again I used the BERT-Tiny model. The only hyperparameter I changed was the weight decay constant for L2 regularization. I tested a weight decay of 0, 0.01, and 0.10. I got roughly identical results with all three choices:

Weight decay	Accuracy
0	0.87
0.01	0.87
0.10	0.87

Accuracy on the training set was similar, suggesting that we avoided overfitting. There was no difference in accuracy between the different emotions.

Conclusion

I greatly enjoyed learning about deep learning techniques in NLP, especially word embeddings. The fact that the dimensions of the word embeddings can capture relationships between words understandable to us is incredible. Understanding the architecture of recurrent neural networks and transformers was challenging but rewarding.

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